# Shuo Sun

Postdoctoral Associate · Singapore-MIT Alliance for Research and Technology (SMART), M3S Interdisciplinary Research Group

He received both his Ph.D. and B.Eng. degrees in Mechanical Engineering from the National University of Singapore, under the supervision of Professor Marcelo H. Ang Jr. His work bridges machine intelligence and human capability through robotics, autonomy, and collective intelligence, with a particular interest in systems that make human-AI collaboration more capable, meaningful, and deployable.

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## APPOINTMENTS

2025 - Present

### Postdoctoral Associate

Singapore-MIT Alliance for Research and Technology (SMART) · M3S Interdisciplinary Research Group · Singapore

Research at the intersection of robotics, autonomous systems, and human-AI collective intelligence.

## EDUCATION

2021 - 2025

### Ph.D. in Mechanical Engineering

National University of Singapore

Research on autonomous systems, motion planning, prediction, and simulation under Professor Marcelo H. Ang Jr.

2016 - 2020

### B.Eng. in Mechanical Engineering

National University of Singapore

Foundation in robotics, engineering design, and interdisciplinary systems work.

## RESEARCH AREAS

ROBOTICS · AUTONOMOUS SYSTEMS · HUMAN-AI COLLECTIVE INTELLIGENCE · MOTION PLANNING AND PREDICTION · GENERATIVE SIMULATION

## SELECTED PROJECTS

2023 - Present

### Autonomous Mini-bus

NUS · Tron-E · Turing Drive · Flagship Deployment

Development of an autonomous mini-bus platform for on-demand shuttle services in geofenced environments.

2020 - 2023

### Autonomous Road Sweeper

NUS · Kärcher Germany · Flagship Deployment

A full autonomous cleaning platform for mixed indoor and outdoor environments, from drive-by-wire retrofitting to full-stack software.

2023 - 2024

### DriveSceneGen

NUS Advanced Robotics Centre · Research System

A generative simulation system that produces diverse and realistic driving scenarios from scratch for autonomous vehicle development and evaluation.

## PUBLICATIONS

conference — ICRA 2026 Late Breaking Results · 2026

### RoboAtlas: Mapping the Global Robotics Landscape

Jiacheng Zhang and Shuo Sun and Vasiliki Charisi and Xinru Wang and Xinyue Chen and Zhexuan Ma and Alok Prakash and Thomas W. Malone

preprint — arXiv · 2026

### Where can AI be used? Insights from a deep ontology of work activities

Alice Cai and Iman YeckehZaare and Shuo Sun and Vasiliki Charisi and Xinru Wang and Aiman Imran and Robert Laubacher and Alok Prakash and Thomas Malone

conference — Advances in Neural Information Processing Systems 38 · 2025

### AGI-Elo: How Far Are We From Mastering A Task?

Shuo Sun and Yiming Zhao and C. D. W. Lee and Jiawei Sun and Chengran Yuan and Zefan Huang and Dongen Li and Justin K. W. Yeoh and Alok Prakash and Thomas Malone journal — IEEE Robotics and Automation Letters · 2025

**IMPACT: Behavioral intention-aware multimodal trajectory prediction with adaptive context trimming**

Jiawei Sun and Xinyu Yue and Jiahui Li and Ting Shen and Chengran Yuan and Shuo Sun and Shitong Guo and Qingwen Zhou and Marcelo H. Ang Jr.

preprint — arXiv · 2025

**PIE: Perception and Interaction Enhanced End-to-End Motion Planning for Autonomous Driving**

Chengran Yuan and Zihan Lu and Zhaozheng Zhang and Yiming Zhao and Zefan Huang and Shuo Sun and Jiawei Sun and Jiahui Li and C. D. W. Lee and Marcelo H. Ang

conference — ICRA 2025 · 2025

**RMP-YOLO: A Robust Motion Predictor for Partially Observable Scenarios even if You Only Look Once**

Jiawei Sun and Jiahui Li and Tingchen Liu and Chengran Yuan and Shuo Sun and Zefan Huang and Anthony Wong and Keng Peng Tee and Marcelo H. Ang

conference — ITSC 2024 · 2024

**ADM: Accelerated diffusion model via estimated priors for robust motion prediction under uncertainties**

Jiahui Li and Ting Shen and Zekai Gu and Jiawei Sun and Chengran Yuan and Yuhang Han and Shuo Sun and Marcelo H. Ang

journal — Nature Communications · 2024

**Computational design of ultra-robust strain sensors for soft robot perception and autonomy**

Haitao Yang and Shuo Ding and Jiahao Wang and Shuo Sun and Ruphan Swaminathan and Serene Wen Ling Ng and Xinglong Pan and Ghim Wei Ho

conference — ITSC 2024 · 2024

**ControlMTR: Control-guided motion transformer with scene-compliant intention points for feasible motion prediction**

Jiawei Sun and Chengran Yuan and Shuo Sun and Sha Wang and Yuhang Han and Shengnan Ma and Zefan Huang and Anthony Wong and Keng Peng Tee and Marcelo H. Ang

preprint — arXiv · 2024

### DRAMA: An efficient end-to-end motion planner for autonomous driving with mamba

Chengran Yuan and Zhaozheng Zhang and Jiawei Sun and Shuo Sun and Zefan Huang and C. D. W. Lee and Dongen Li and Yuhang Han and Anthony Wong and Keng Peng Tee and Marcelo H. Ang

journal IEEE Robotics and Automation Letters · 2024

### DriveSceneGen: Generating Diverse and Realistic Driving Scenarios From Scratch

Shuo Sun and Zekai Gu and Tianchen Sun and Jiawei Sun and Chengran Yuan and Yuhang Han and Dongen Li and Marcelo H. Ang Jr.

conference IEEE Conference on Cybernetics and Intelligent Systems 2023 · 2023

### An Adaptive Local Context Extraction Method for Motion Prediction and Planning

Jiawei Sun and Ting Liu and Chengran Yuan and Shuo Sun and Anthony Wong and Keng Peng Tee and Marcelo H. Ang

preprint arXiv · 2023

### CARLA-Loc: synthetic SLAM dataset with full-stack sensor setup in challenging weather and dynamic environments

Yuhang Han and Zhiyang Liu and Shuo Sun and Dongen Li and Jiawei Sun and Chengran Yuan and Marcelo H. Ang Jr.

conference IROS 2023 · 2023

### FISS+: Efficient and Focused Trajectory Generation and Refinement using Fast Iterative Search and Sampling Strategy

Shuo Sun and Jie Chen and Jiawei Sun and Chengran Yuan and Yuanchen Li and Tangyike Zhang and Marcelo H. Ang

conference ICCCR 2023 · 2023

### GET-DIPP: Graph-embedded transformer for differentiable integrated prediction and planning

Jiawei Sun and Chengran Yuan and Shuo Sun and Zhiyang Liu and Tiong Goh and Anthony Wong and Keng Peng Tee and Marcelo H. Ang

conference ITSC 2023 · 2023

### Maximum Entropy Inverse Reinforcement Learning Based on Frenet Frame Sampling for Human-like Autonomous Driving

Tangyike Zhang and Shuo Sun and Jiamin Shi and Shitao Chen and Marcelo H. Ang and Jingmin Xin and Nanning Zheng conference ICARCV 2022 · 2022

### Autonomous Research Platform for Cleaning Operations in Mixed Indoor & Outdoor Environments

Shuo Sun and Tangyike Zhang and Zhihai Xiang and Yuhang Han and Dongen Li and Jianghao Li and Zhiyang Liu and Marcelo H. Ang

conference ITSC 2022 · 2022

### Exploring the effect of 3D object removal using deep learning for LiDAR-based mapping and long-term vehicular localization

Yiming Chen and Shuo Sun and Huan Yin and Marcelo H. Ang

journal IEEE Robotics and Automation Letters · 2022

### FISS: A Trajectory Planning Framework Using Fast Iterative Search and Sampling Strategy for Autonomous Driving

Shuo Sun and Zhiyang Liu and Huan Yin and Marcelo H. Ang

## MENTORING

2019 - Present

### Student supervision and mentoring

Mentored more than 70 student theses, capstone, and research projects across robotics and autonomous systems.